

# SOVIN BIRLA

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## SUMMARY

Seasoned Backend Engineer with 4+ years of experience building production-grade services for high-scale SaaS and healthcare platforms. Skilled in designing and scaling APIs, modeling multi task systems, and integrating complex data stores like PostgreSQL, MongoDB, Redis, and Elasticsearch. Expertise in GCP and AWS infrastructure, CI/CD, Kubernetes, and PyTorch. Proven ability to lead distributed teams and deliver high-quality backend architecture and integrations.

## EDUCATION

**Masters in Applied Data Science**

**University of Chicago**

Mar 2025 - Mar 2026

**Visiting Graduate Student**

**Stanford University**

Jan 2024 - July 2024

**BS Statistics + Computer Science**

**University of Illinois Urbana-Champaign**

Aug 2017 - May 2021

## EXPERIENCE

**Machine Learning Intern, Elevance Health, Remote / Atlanta, GA**

May 2024 - Jul 2024

- Developed a real-time ML-integrated medical chatbot using FastAPI, AWS, Kubernetes, and Ollama enabling text-to-3D lung reconstruction pipeline with vision transformers and diffusion models
- Created high-resolution 3D lung models with 3D Slicer's REST API and MONAI's from the realistic generated CT scans provided by the text prompts
- Expanded the training dataset by 40% using fine-tuned LLMs (Phi-3, Mistral 7B) and BiomedCLIP to improve automated CT scan analysis.
- Built a stroke prediction model with 81% accuracy using feature engineering and Random Forest/AdaBoost classifiers for early risk assessment.

**Software Engineer, Maven Wave (Google Cloud Consultancy), Chicago, IL**

Jun 2021- Dec 2023

- Designed and deployed 5+ production-ready backend systems for clients including EY, McDonald's, Capital One, and United Health Group using Django, FastAPI, and cloud-native tools.
- Built scalable APIs and asynchronous task pipelines using RabbitMQ and Docker in AWS and GCP.
- Managed infrastructure with Kubernetes, Terraform, and Jenkins CI/CD, improving release cycles by 25%.
- Led an 8-person international team using Agile methodologies, sprint planning, and ticket tracking (Trello, JIRA).
- Worked closely with ML teams to integrate inference services into backend applications.

## PROJECTS

**Distributed Raspberry Pi SLURM Cluster**

- Set up a mini HPC cluster with 4 Raspberry Pi nodes to run and schedule compute-intensive inference batch jobs on LLM's using fine tuning techniques like LoRA
- Implemented multithreaded task execution in C++ and configured shared storage across nodes

**Ophthalmology Research Platform (Optha-Llama2)**

- Fine-tuned a healthcare-specific LLM for biomedical causal extraction using BEL (Biological Expression Language).
- Deployed the model via RayServe + Argoflow and participated in Stanford HCAI discussions on AI safety.

## SKILLS

- **Languages:** C, C++, Python, Java
- **Databases:** MySQL, PostgreSQL, MongoDB, Oracle SQL
- **Frameworks:** FastAPI, Django, React, Keras, Matplotlib
- **Cloud & DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Firebase, Helm, SLURM